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Mahidol University



Doctoral Seminar Presentation

Learning Sparse Feature-Interaction Graphs with Attention-Based GNNs

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Abstract

This paper studies whether an attention-based Graph Neural Network (GNN) can capture interaction groups and select a sparse feature graph on the dataset containing interactions with known ground truth. Features are treated as nodes in a complete graph; a one-layer TransformerConv is trained to obtain attention coefficients, which are aggregated into a symmetric, min-max-normalized edge score (MF). A single global threshold τ selects edges, and the predictor is retrained on the pruned graph.

On a synthetic dataset with higher-order interactions, the method highlights the true groups: with $\tau=0.68$, it selects 9 edges (vs. 45 in the complete graph; 8 in the ground truth) and achieves Precision 0.89, Recall 1.00, and $F_1=0.94$ for edge recovery. Predictive accuracy is preserved or improved: MAE is 0.072 for the pruned graph, compared with 0.088 on the complete graph, 0.070 for a ground-truth oracle, and 0.397 for a null (no-edge) graph. These results suggest that removing noisy edges yields a compact, interpretable graph without harming prediction. The procedure is generic and can be tested on any synthetic dataset that provides ground-truth interactions for evaluation.

Keywords: Feature Graph, Feature Interaction, Graph Neural Network

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